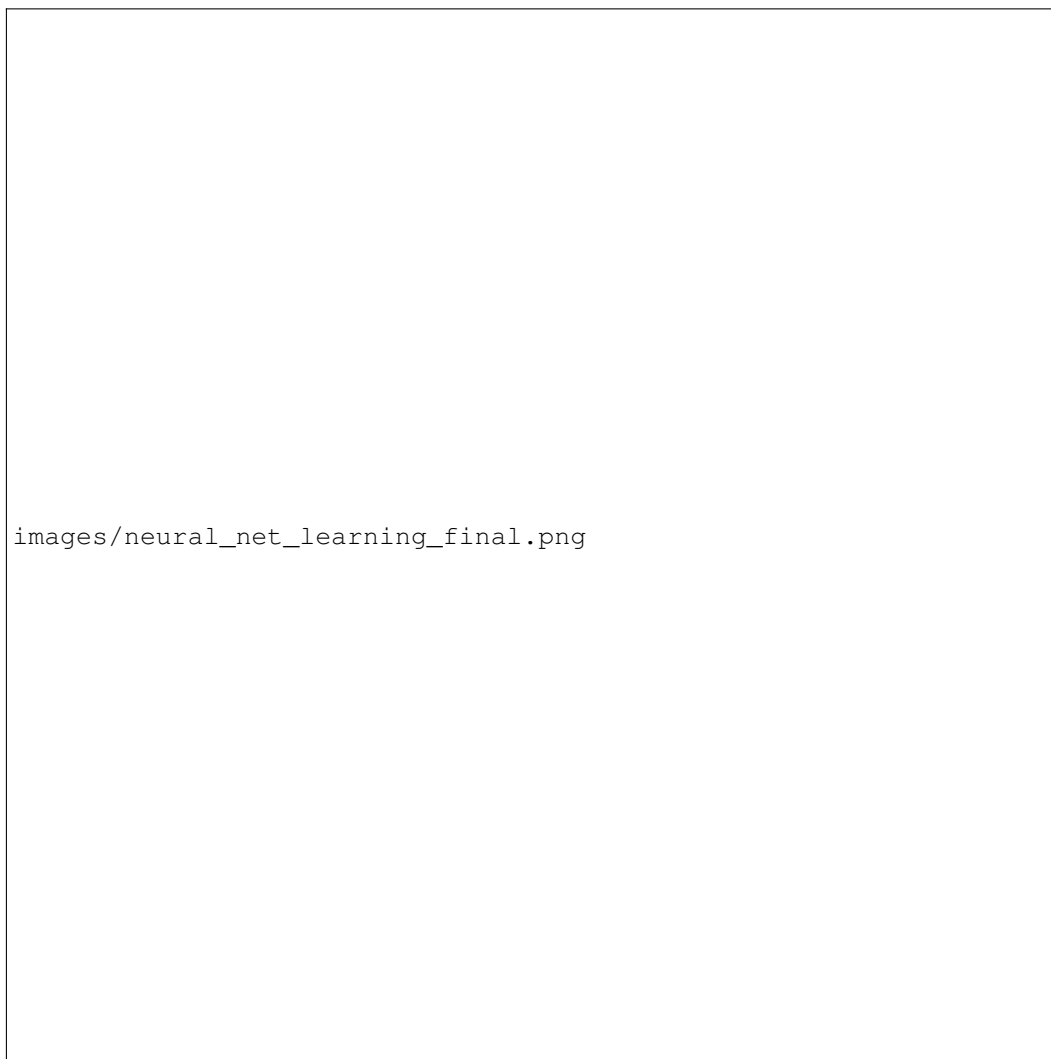


grants #1545126 and #1911835, and grant #61454 from the John Templeton Foundation.

Ethics Statement

Equipping artificial agents with the capacity to learn from linguistic feedback is an important step towards value alignment between humans and machines, with the end goal of supporting beneficial interactions. However, one risk is expanding the set of roles that such agents can play to those requiring significant interaction with humans – roles currently restricted to human agents. As a consequence, certain jobs may be more readily replaced by artificial agents. On the other hand, being able to provide verbal feedback to such agents could expand the group of people able to interact with them, creating new opportunities for people with disabilities or less formal training in computer science.

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images/neural_net_learning_final.png

Figure 4: Left: A trajectory from our experiment. Right: Inference network output given this trajectory. Top row: the model learns to map feature-related tokens (“Descriptive” feedback) directly to rewards, independent of the trajectory. Bottom left / center: the model maps praise and criticism (“Evaluative” feedback) through the feature-counts from the trajectory. Bottom right: a failure mode. “Descriptive” feedback with negative sentiment, a rare speech pattern, is not handled correctly.